

RiverEye 新店溪望愛 — 互動式災情通報地圖平台

System Plan

Revision 2

Revision History

Revision	Description
Revision 2	Text corrections. Added message IDs for message texts. Updated the specification to align with Next.js, v0 (Vercel AI UI Generator), Google Maps JavaScript API event listeners, and real-time structured geospatial data from the Water Resources Agency (WRA) RESTful API.
Revision 1	Initial Revision. Base framework definition.

1. System Specification

1.1 Requirement Definition

RiverEye is web-based and features the use of (1) open source software, (2) ISO/TC211 and OGC standards to process related meta data, (3) coordinate systems used in Taiwan (TWD97 / TM2), (4) WMS or WMTS for sharing geospatial services for the existing systems.

1.2 Development Environment

- **Target web browsers:** Google Chrome, Microsoft Edge, Apple Safari.
- **Development tools:** Next.js,v0 (Vercel AI UI Generator), Supabase Cloud (PostgreSQL / PostGIS).
- **Web-based GIS map view:**
 - **Google Maps JavaScript API:** The core map engine of this system. It is utilized to render high-precision maps, Advanced Markers, and InfoWindows on the web client, while also handling real-time user geolocation to deliver a seamless, interactive map experience.
 - **WRA (Water Resources Agency) RESTful API:** Used for fetching real-time, structured geospatial data (in JSON format) from various observation stations across Taiwan—including rainfall, river water levels, and flooding sensors. This data is transmitted via the Next.js backend to the frontend map for dynamic rendering.

1.3 Data Standard

- Follow ISO/TC211 and OGC standards to process related metadata.
- Conform to the coordinate systems in Taiwan (TWD97 2-degree TM, EPSG:3826 / EPSG:4326).

2. System Scope

RiverEye includes 3 subsystems as below:

- **System A: Client-Side Reporting Subsystem** (民眾端災情即時回報子系統) - Provides frontend map interface for citizens to pinpoint location coordinates, select hazard types (flood, mudslide, road closed), and submit real-time reports.
- **System B: Admin Management & Dispatch Subsystem** (管理員後端審查與調度子系統) - Enables administrators to review, update status (Pending, In Progress, Resolved), and optimize emergency resource allocation via a dashboard.
- **System C: Spatial Data & Watershed Integration Subsystem** (空間圖資與水源特定區整合子系統) - Integrates the Water Resources Agency (WRA) specific area map layers and watershed boundaries to provide high-precision flood prevention reference data for citizen reporting and administrative dispatch.

2.1 Subsystem Relationship Diagram

[System A: Client Report] --(Real-time Data via Supabase)--> [System B: Admin Dashboard]
[System C: WRA Map Layers / Watershed Boundary] --(OGC Standards)--> [System A & System B Map Views]

3. Map Layer Architecture (M0)

The system initializes the interactive map interface with specific layers that can be toggled by the user or loaded by the OS.

Available Independent Layers:

Layer Name (Layer ID)	Description
水利署即時觀測圖層	WRA open data overlaid with the Xindian River Special Zone watershed provides a key spatial reference for flood prevention and crowdsourcing.

Available Mutually Exclusive Layers (Exactly one can be active):

Layer Name	Description
標準地形圖	Standard street map displaying topographic features and contours.
混合衛星圖	Standard street map + Satellite map (Hybrid street map).

4. Map Query (M0-F)

This view appears when the user taps the M0-F button in M0. Upon setting the search criteria and pressing **M0-F04**, the app will retrieve query results from the web service and display them as a 特定區流域範圍 marker (**M0-F02**) on M0-M.

Element ID	UI Element Name	Description / Specification
M0-F01	目前位置	This marker marks the user's current GPS location on the map using Geolocation API.
M0-F02	座標定位	The watershed specific area marker will contain an icon and a text annotation showing the current location at that marker point.
M0-F03	未回報災害狀態標記	This marker denotes that a report for the specific location has not been filed yet.

M0-F04	空間查詢執行按鈕	A front-end action button (corresponding to the original template F0-03) that, when clicked, sends a request to the spatial Web Service and returns watershed protection area information.
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5. User Authorization & Login (L0)

The login view is the first view the user sees when the app is started for the first time, or if the user had logged out. During login, the server checks the user credentials via Supabase Auth, as well as determines the permissions (Citizen vs. Admin) for the user.

UI Elements:

Element ID	UI Element Name	Specification / Validation Rules
L0-01	Title of the App	Displays "RiverEye 新店溪望愛 — 互動式災情通報平台"
L0-02	Username edit box	Email input field. e.g., lily940301@gmail.com (Must conform to email string pattern).
L0-03	Password edit box	The password will be censored on screen (Input type="password").
L0-04	Error message	Dynamic text area displaying error texts such as L0-M02.
L0-05	Login button	Triggers authentication request to Supabase Auth backend.

System Messages & Error Identifiers:

Message ID	Message Text (Traditional Chinese)	Trigger Condition
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L0-M01	欄位不可為空，請輸入帳號與密碼。	Triggered when L0-05 is clicked but L0-02 or L0-03 is empty.
L0-M02	使用者帳號或密碼錯誤，請重新輸入。	Triggered when backend auth fails (Invalid credentials).
L0-M03	網路連線異常，請檢查您的網路設定。	Triggered when connection to Supabase server times out.